Counterfactual Action Landscapes for PPO

A critical pitch memo on why the injection interface matters, and what the current four-environment results support.

One-line pitch

Counterfactual advantages are not plug-and-play in PPO: adding them before clipping and adding them as a separate loss coincide only in PPO's locally linear regime, and this distinction explains why all-action CF losses, sampled perturbations, and distilled perturbations produce different learning behavior.

Why this is important

Counterfactual credit assignment is already well established. The useful pitch is narrower: once a counterfactual action landscape is available, modern clipped policy optimization still has an unresolved interface problem. PPO is not a raw linear policy-gradient estimator; clipping, normalization, minibatching, and gradient-scale effects decide whether the same CF signal helps, washes out, or changes the effective trust region.

This reframes the contribution from 'we use counterfactuals' to 'we identify and test optimizer-facing interfaces for counterfactual credit in clipped PPO.' That is the part reviewers are less likely to dismiss as already known.

Core mathematical distinction

Let r_t(theta) = pi_theta(a_t|s_t) / pi_old(a_t|s_t). Define the policy-centered counterfactual advantage:

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A_CF(s_t,a) = Q_CF(s_t,a) - sum_b pi_old(b|s_t) Q_CF(s_t,b)
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Advantage perturbation forms a single scalar advantage before PPO clipping:

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A_tilde_t = A_GAE_t + beta A_CF(s_t,a_t)
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and optimizes:

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L_perturb(theta) = -E_t[min(r_t A_tilde_t, clip(r_t,1-eps,1+eps) A_tilde_t)]
```

If clipping is inactive, or if both terms remain on the same clipping branch, this behaves approximately like L_GAE + beta L_CF. But in real PPO the clipping map is nonlinear, so generally:

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L_PPO(A_GAE + beta A_CF) != L_PPO(A_GAE) + beta L_PPO(A_CF)
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The perturbation can change the sign, magnitude, and active clipping branch of the PPO update. This is the central pitch: the counterfactual estimator and the PPO interface cannot be evaluated independently.

Current empirical readout

Environment	Best from current curves	Worst	Interpretation
RedBlueDoors-6x6	cf_all_action	gae	Queried perturbation learns earlier, but all-action CF finishes clearly highest.
UnlockPickup	cf_queried_perturb	gae	Perturbation is strongly sample-efficient; all methods converge near similar final return.
Taxi-v4	cf_all_action	landscape_distill	All-action CF has best final return; distillation is weak here.

Environment	Best from current curves	Worst	Interpretation
DoorKey-6x6	cf_all_action	gae	All-action CF is faster and slightly higher final; both perturbation and distillation beat or match GAE.

Conservative claim: across these four environments, counterfactual information usually helps over vanilla GAE, but the best injection mechanism changes by task and metric. All-action CF is strongest overall for final performance; queried perturbation can be better for early learning.

What this supports - and what it does not

Claim	Status	Reason
CF perturbation improves over vanilla GAE	Supported cautiously	It matches or beats GAE in the shown curves, especially UnlockPickup and RedBlueDoors.
All-action CF is best overall	Mostly supported	It wins final performance on RedBlueDoors, Taxi, and DoorKey; UnlockPickup favors perturbation for speed.
Advantage perturbation equals a separate CF loss	Not supported	They coincide only in PPO's locally linear regime; clipping breaks additivity.
Distillation solves the oracle cost	Not yet	Mixed results: useful on some MiniGrid tasks, weak on Taxi and RedBlueDoors.
PPO-CF is a universal new SOTA algorithm	Do not claim	The evidence is too small, too mixed, and not yet compute-normalized.

ICLR-facing narrative

Problem	Counterfactual action landscapes provide richer per-state credit than sampled trajectories, but PPO's clipped objective creates multiple non-equivalent ways to use them.
Hypothesis	The optimizer-facing interface is as important as the counterfactual estimator.
Method	Compare all-action CF surrogate, queried sampled advantage perturbation, and learned/distilled perturbation under matched environments.
Finding	All-action CF gives the most consistent final gains; sampled perturbation can improve early learning; distillation is promising but unreliable.
Message	Counterfactual credit is useful, but not plug-and-play. In PPO, where the signal enters the clipping operation changes the algorithm.

What must be added before submission

Needed evidence	Why reviewers will ask for it
Multiple seeds with confidence intervals	The current curves alone are not statistically enough.
AUC, time-to-threshold, and final return	Different algorithms win on different metrics.
Compute-normalized comparison	All-action CF is expensive; sample efficiency alone is not the whole cost.
Matched KL or target-KL runs	Rules out the possibility that CF simply changes effective step size.
Shuffled CF control	Tests whether the gain is state-local counterfactual information rather than extra gradient structure.
Beta/alpha sweeps	Avoids a hyperparameter cherry-picking criticism.
Clipping diagnostics	This is central to the pitch, so measure branch changes and clip fraction changes directly.

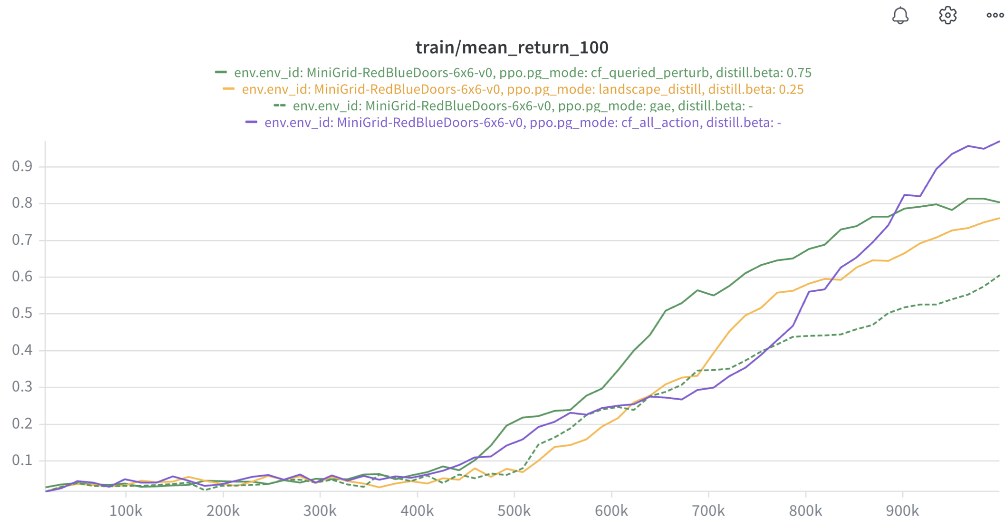
Recommended paper title options

Option	Comment
Counterfactual Action Landscapes for Clipped Policy Optimization	Best technical framing.
Where Should Counterfactual Credit Enter PPO?	Best question-driven framing.
Counterfactual Credit Is Not Plug-and-Play in PPO	Best provocative framing, but maybe less formal.

Attached current curves

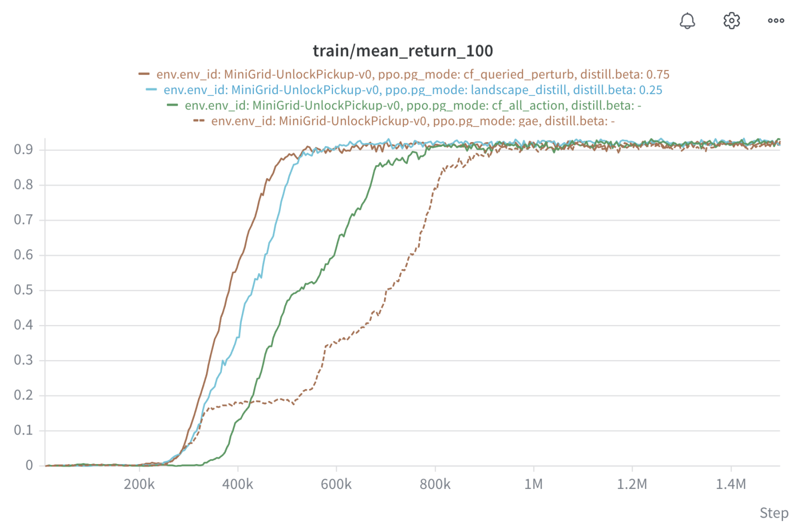
These figures are the current visual evidence from the four environments. They should be treated as provisional until backed by seed-level confidence intervals and scalar summary metrics.

RedBlueDoors-6x6



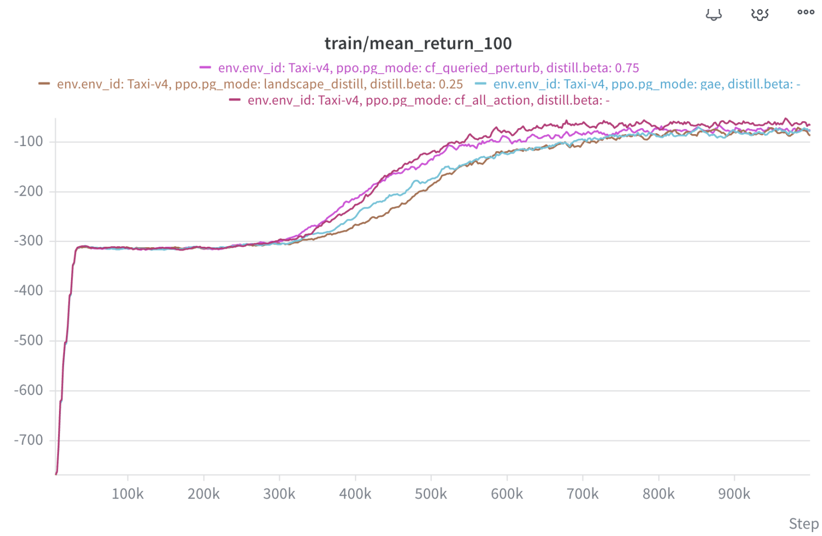
All-action CF reaches the strongest final return; queried perturbation is faster early, but is overtaken.

UnlockPickup



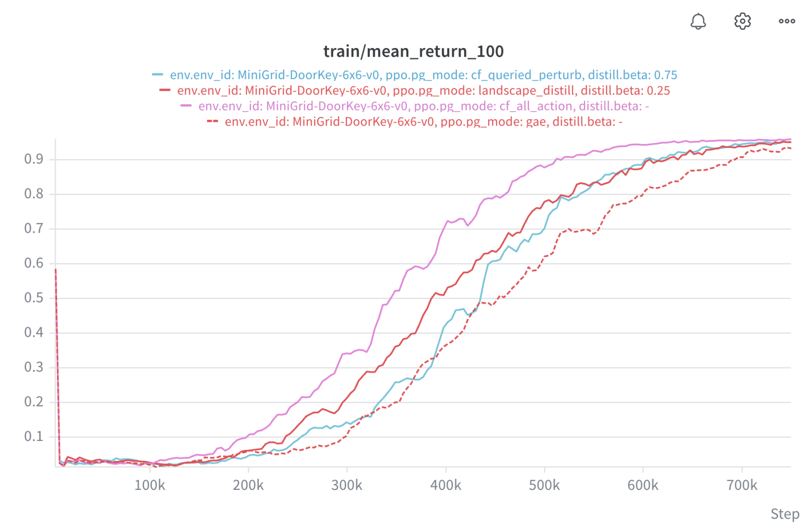
Queried perturbation is the most sample-efficient; all methods eventually cluster near the same final return.

Taxi-v4



All-action CF has the best final return; queried perturbation helps; distillation is not reliable here.

DoorKey-6x6



All-action CF is clearly faster and slightly higher final; distillation and queried perturbation beat or match GAE.

Bottom line

The strongest pitch is not that PPO-CF universally beats PPO. The strongest pitch is that counterfactual action information changes PPO in interface-dependent ways. In the locally linear regime, perturbing the advantage can look like adding a CF loss. Under clipping, it becomes a distinct algorithm. Your current results are confusing only under the old framing; under this framing, the confusion is the evidence.

Recommended claim: counterfactual landscapes provide useful policy-gradient information, but in PPO the clipping interface determines whether that information appears as variance reduction, advantage correction, step-size change, or an unstable/distilled guide.